

Module 04

Dockerfile and Registry

DevOps end to end services & solutions

Agenda

- > Dockerfile – Basic commands
- > Dockerfile – Based on node:14 OS
- > Docker Build and Docker Run Image
- > Multi-Stage Build
- > Docker Registry – Tag and Push Image
- > Hands-On – Lab-04 – Dockerfile and Registry

What is a Dockerfile?



Blueprint

A script containing a set of instructions to automate the building of Docker images.



Setup

Defines everything needed to set up an app inside a container (OS, code, dependencies).



Automation

Contains all the commands a user could call on the command line to assemble an image.

Dockerfile – Basic Commands

- > **FROM:** Specifies the base image.
- > **WORKDIR:** Sets the working directory inside the container.
- > **COPY & ADD:** Copies files from host to container.
- > **RUN:** Executes commands during build time.
- > **CMD & ENTRYPOINT:** Defines the default command when running the container.
- > **EXPOSE:** Opens a port for the container.

```
FROM ubuntu:22.04
```

```
RUN apt-get update -y
```

```
RUN apt-get upgrade -y
```

```
RUN apt-get install -y curl
```

Dockerfile – Based on node:14 OS

```
# Use an official Node.js runtime as a parent image
FROM node:14
# Set the working directory in the container
WORKDIR /usr/src/app
# Copy package.json and package-lock.json to the working directory
COPY package*.json ./
# Install any needed dependencies
RUN npm install
# Copy the rest of the application code to the working directory
COPY . .
# Make the container's port 3000 available to the outside
EXPOSE 3000
# Define the command to run the application
CMD ["npm", "start"]
```

Docker Build and Run

Docker Build

Creates an image from a Dockerfile.
Packages your code to be shipped
anywhere.

```
docker build -t node-app .
```

Docker Run

Creates a running container from an
image. Use `-d` for background mode.

```
docker run -p 3000:3000 node-app  
# Or run in background  
docker run -d -p 3000:3000 node-app
```

Multi-Stage Build

Multi-stage builds allow you to drastically reduce the size of your final image by using multiple FROM instructions.

You can copy artifacts from one stage to another, leaving behind all the build tools and intermediate files.

```
1  # Stage - Development
2  FROM Dev-Image as dev
3  . . .
4  . . .
5
6  # Stage - Testing
7  FROM Test-Image as test
8  COPY --from=dev /src/app .
9  . . .
10 . . .
11
12 # Stage - Production
13 FROM Minimum-Image as prod
14 COPY --from=dev /src/app .
15 . . .
16 . . .
```

stage 0

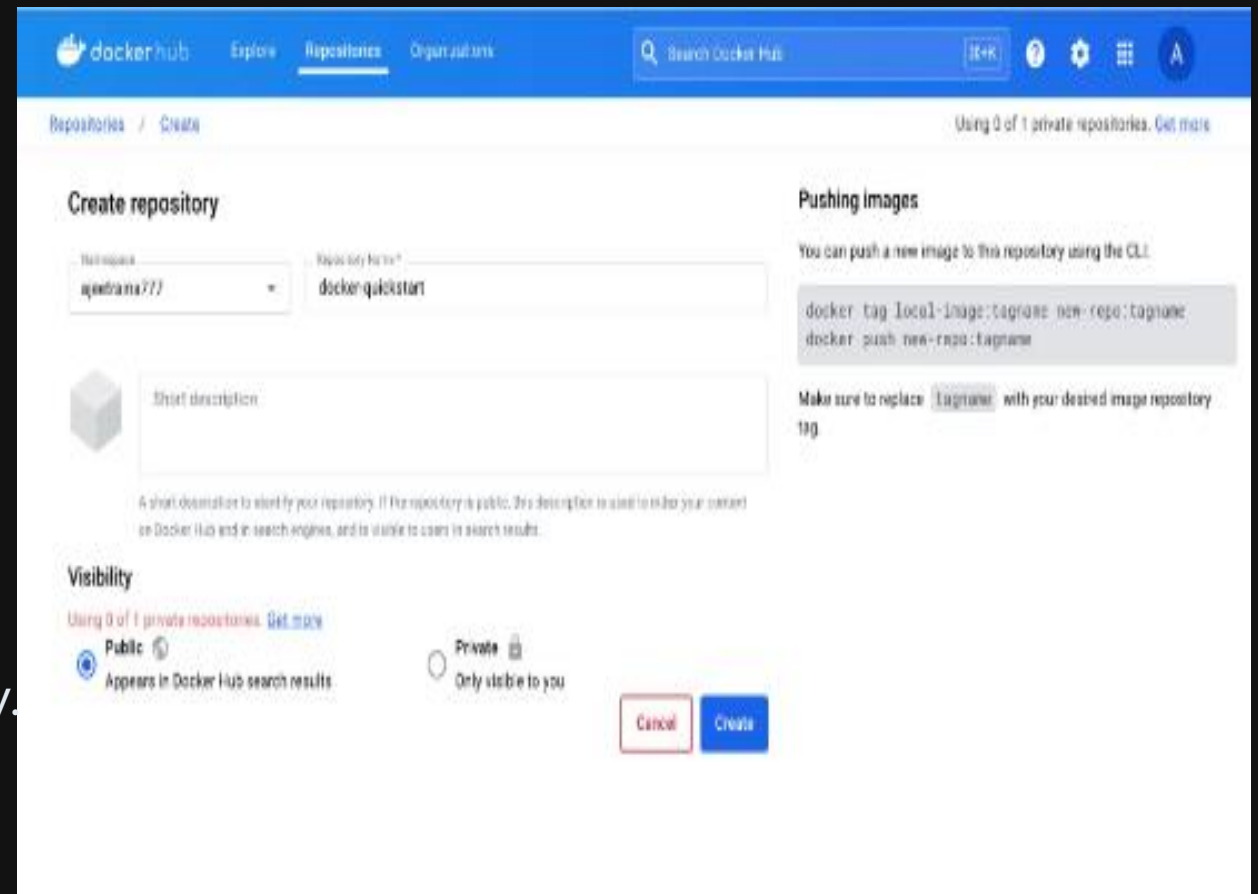
stage 1

stage 2

Docker Registry – Tag and Push

Steps to Publish

- > **Login:** Authenticate with Docker Hub.
- > **Tag:** Rename your local image to match your registry repository.
- > **Push:** Upload the image to the registry.



The screenshot shows the Docker Hub 'Create repository' page. The top navigation bar includes 'docker hub', 'Explore', 'Repositories', and 'Organizations'. A search bar is on the right. The main heading is 'Create repository'. Below it, there are two input fields: 'Repository name' with the value 'agorhans777' and 'Repository type' with the value 'docker-quickstart'. A 'Short description' text area is below these. A note explains that the description is used for search results. The 'Visibility' section shows 'Public' selected, with a note 'Appears in Docker Hub search results'. The 'Private' option is also visible. On the right, a 'Pushing images' section shows CLI commands: 'docker tag local-image:tagname new-repo:tagname' and 'docker push new-repo:tagname'. At the bottom right are 'Cancel' and 'Create' buttons.

Tag & Push Commands

1. Login to Docker Hub

docker login

2. Tag the image

docker tag /:

docker tag myapp mydockeruser/myapp:latest

3. Push the image

docker push mydockeruser/myapp:latest

4. Pull (Verify)

docker pull mydockeruser/myapp:latest

Hands-On – Lab 04: Dinner Suggestion

- > Create a app.py application (Python Flask).
- > Wrap it with a Dockerfile.
- > Build the image.
- > Run the image and access via browser.
- > Test via interactive mode.
- > Tag and Push to Docker Hub.



Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

